Session 0: Introduction to the Course

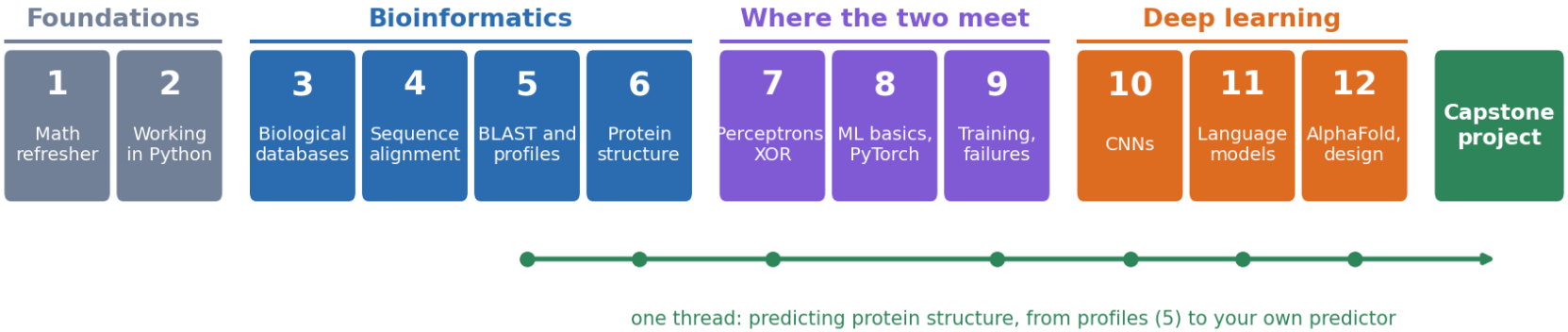
KB8029 — Bioinformatics & Deep Learning

Arne Elofsson and Petras Kundrotas

SciLifeLab

SciLifeLab

Bioinformatics and machine learning, taught together



Every session: a page in this book, a notebook, a lecture with live quizzes and discussion notebooks, and (most sessions) a lab submitted in Canvas.

Sessions 1-2: the maths and Python we need. 3-6: classical bioinformatics and protein structure. 7-9: the two meet. 10-12: modern deep learning, up to AlphaFold. One thread, predicting protein structure, ends in your capstone project.

See course.bioinfo.se, 'Session 0: Introduction to the Course'.

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Open, new, and not perfect: tell us

- **All pages, notebooks and labs are open access**
- New this year: there will be errors and unclear parts
- **Found one? Tell us, we appreciate it**
- Feedback form, with your university login; anonymous if you like
- Or a GitHub issue (public), or email arne@bioinfo.se



Feedback form: bioinfo.se/internal/kb8029-feedback (also linked at the bottom of every course page) https://bioinfo.se/internal/kb8029-feedback/

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By the end of the course you can

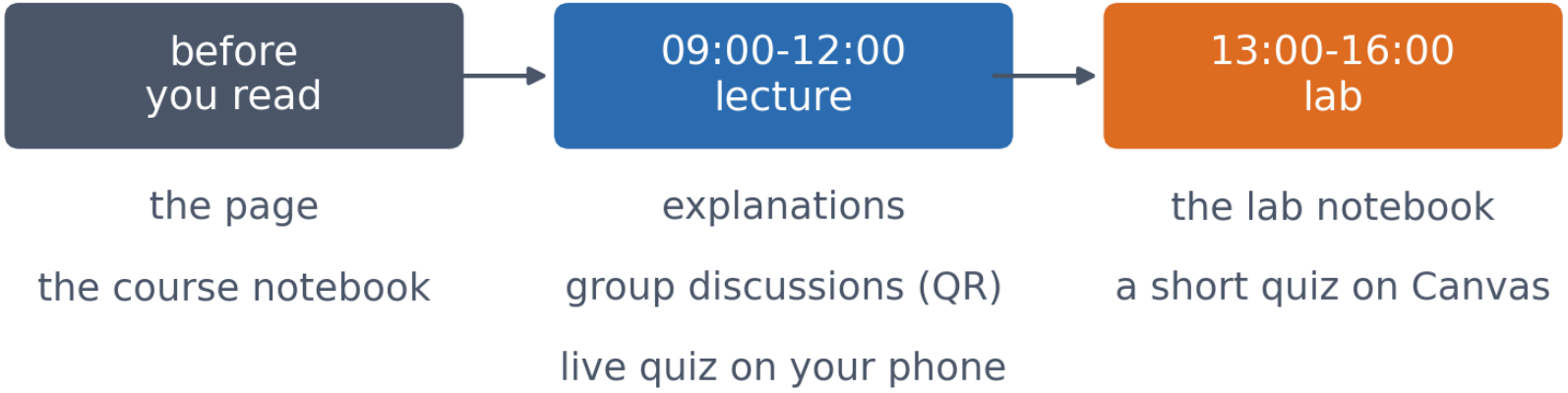
Find and compare sequences and structures databases and tools	Sessions 3-6
Explain the classical methods alignment, substitution matrices, profiles	Sessions 4-5
Build, train and evaluate neural networks in PyTorch; is the result real?	Sessions 7-9
Explain modern deep learning for proteins CNNs, language models, AlphaFold	Sessions 10-12
Design your own predictor secondary structure	capstone project

The last one is the capstone project: your own secondary-structure predictor.

Full plan: course.bioinfo.se/schedule.html https://course.bioinfo.se/schedule.html

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Every session has the same parts



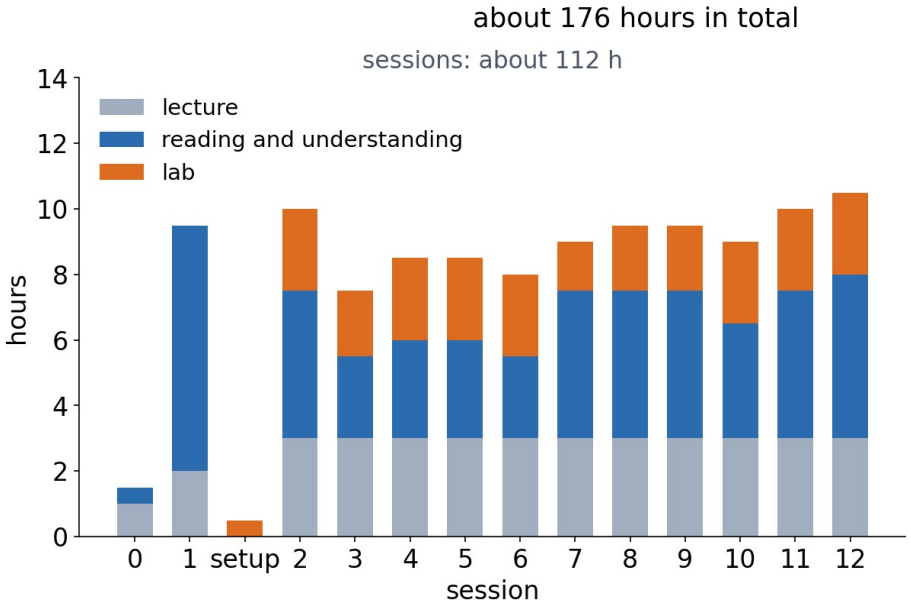
Lectures are not recorded: read the page first.

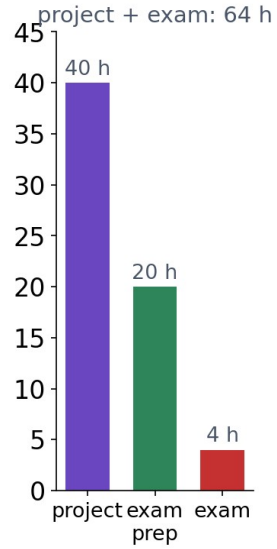
The page and its course notebook (with an 'Open in Colab' badge) before the lecture; the lab notebook (opened in VS Code) and its Canvas quiz in the afternoon.

See 'How a session works' on the Session 0 page.

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About 176 hours: four weeks and two days, full time





- **Each session: lecture, reading, lab (8-10.5 h)**
- Lab: fits the 3-hour afternoon
- Project 40 h, exam preparation 20 h, exam 4 h

See course.bioinfo.se/course-info.html#workload, https://course.bioinfo.se/course-info.html#workload

SciLifeLab

Where to find everything

course.bioinfo.se	canvas.su.se	quiz.bioinfo.se
all pages and notebooks	dates, rooms, deadlines	live quizzes
the whole book as one PDF	lab quizzes	during the lectures
setup, rules, schedule	discussion forum	on your phone

Questions: the Canvas forum, so everyone sees the answer

The Setup page (VS Code and Colab) and the Course Information page (grading, lab rules, plagiarism) are on course.bioinfo.se.

Links on the Session 0 page, 'Where to find everything'. https://course.bioinfo.se/

Teachers and teaching assistants

Teachers



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Teaching assistants: labs and the project presentations



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Course questions: the Canvas forum, so that everyone sees the answer. Personal matters: email a teacher. Lab help: during the lab hours.

Contact details also on Canvas, 'Teachers and contact'. Drawn from tools/canvas/people.csv.

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Schedule, part 1: lectures 09-12, labs 13-16

2027-02-22 to 2027-03-05

date	time	what	teacher	room
Mon 22 Feb	09:00-12:00	Lecture 0+1: Introduction + ...	Arne	Air&Fire, Gamma 2
Wed 24 Feb	09:00-12:00	Lecture 2: Working like a ...	Arne	Air&Fire, Gamma 2
Wed 24 Feb	13:00-16:00	Lab 2: Working like a ...	Arne + Matteo, Vsevolod	Air&Fire, Gamma 2
Fri 26 Feb	09:00-12:00	Lecture 3: Biological Databases	Petras	Air&Fire, Gamma 2
Fri 26 Feb	13:00-16:00	Lab 3: Biological Databases	Petras + Marko, Matteo	Air&Fire, Gamma 2
Mon 01 Mar	09:00-12:00	Lecture 4: Sequence Alignment	Petras	Air&Fire, Gamma 2
Mon 01 Mar	13:00-16:00	Lab 4: Sequence Alignment	Petras + Marko, Matteo	Air&Fire, Gamma 2
Wed 03 Mar	09:00-12:00	Lecture 5: Advanced Database ...	Petras	Air&Fire, Gamma 2
Wed 03 Mar	13:00-16:00	Lab 5: Advanced Database Search ...	Petras + Sarah, Marko	Air&Fire, Gamma 2
Fri 05 Mar	09:00-12:00	Lecture 6: Protein Structure	Petras	Air&Fire, Gamma 2
Fri 05 Mar	13:00-16:00	Lab 6: Protein Structure	Petras + Sarah, Marko	Air&Fire, Gamma 2

The current plan, including who teaches what; the Canvas calendar and Schedule page always have the latest version.

Drawn from tools/canvas/schedule.csv, the same file as the Canvas calendar.

SciLifeLab

Schedule, part 2

2027-03-08 to 2027-03-16

date	time	what	teacher	room
Mon 08 Mar	09:00-12:00	Lecture 7: Linear Models ...	Arne	Air&Fire, Gamma 2
Mon 08 Mar	13:00-16:00	Lab 7: Linear Models, Gradient ...	Arne + Vsevolod, Ines	Air&Fire, Gamma 2
Tue 09 Mar	09:00-12:00	Lecture 8: Machine Learning ...	Arne	Air&Fire, Gamma 2
Wed 10 Mar	09:00-10:00	Capstone: introduction	Marko	Air&Fire
Wed 10 Mar	10:00-12:00	Q&A: Sessions 1-6	Petras	Air&Fire
Wed 10 Mar	14:00-17:00	Lab 8: Machine Learning ...	Arne + Ines, Vsevolod	Lunch Room, Gamma 2
Fri 12 Mar	09:00-12:00	Lecture 9: How a Neural Network ...	Arne	Milky Way & Andr., Alfa 2
Fri 12 Mar	14:00-17:00	Lab 9: How a Neural Network ...	Arne + Vsevolod, Ines	Lunch Room, Gamma 2
Mon 15 Mar	09:00-12:00	Lecture 10: CNNs, from ImageNet ...	Arne	Air&Fire, Gamma 2
Mon 15 Mar	13:00-16:00	Lab 10: CNNs, from ImageNet to ...	Arne + Ines, Sarah	Air&Fire, Gamma 2
Tue 16 Mar	09:00-12:00	Lecture 11: Sequences, Language ...	Arne	Air&Fire, Gamma 2

Not every week follows the Monday-Wednesday-Friday pattern, and some labs and lectures are in other rooms.

Drawn from tools/canvas/schedule.csv, the same file as the Canvas calendar.

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Schedule, part 3: the exam and the project

2027-03-17 to 2027-03-25

date	time	what	teacher	room
Wed 17 Mar	13:00-16:00	Lab 11: Sequences, Language …	Arne + Ines, Vsevolod	Air&Fire, Gamma 2
Thu 18 Mar	13:00-16:00	Capstone: help	Sarah	Milky Way & Andr., Alfa 2
Fri 19 Mar	09:00-12:00	Lecture 12: AlphaFold and …	Arne	Air&Fire, Gamma 2
Fri 19 Mar	13:00-15:00	Q&A: Sessions 7-12	Arne	Air&Fire, Gamma 2
Fri 19 Mar	15:00-18:00	Lab 12: AlphaFold and …	Arne + Matteo, Sarah	Air&Fire, Gamma 2
Mon 22 Mar	08:30-12:30	Written exam		Air&Fire, Gamma 2
Mon 22 Mar	13:00-17:00	Capstone: orals (early slot)	all TAs	Madrid + London, Gamma 7
Wed 24 Mar	09:00-17:00	Capstone: orals (regular time)	all TAs	Earth + Water, Gamma 2
Thu 25 Mar	23:59	Capstone: code and report	all TAs	Online

The teaching assistants of each lab are added to the schedule when decided.

Drawn from tools/canvas/schedule.csv, the same file as the Canvas calendar.

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Getting to SciLifeLab

SciLifeLab

Tomtebodavägen 23A, 171 65 Solna

bus Bus 3 (or others) to Tomteboda-Karolinska inst. (Solna)

rooms
Gamma 7: London, Madrid
Gamma 2: Air&Fire, Earth, Lunch Room, Water
Alfa 2: Milky Way and Andromeda

offices Alfa, floor 6

access your SciLifeLab card at the speed gates

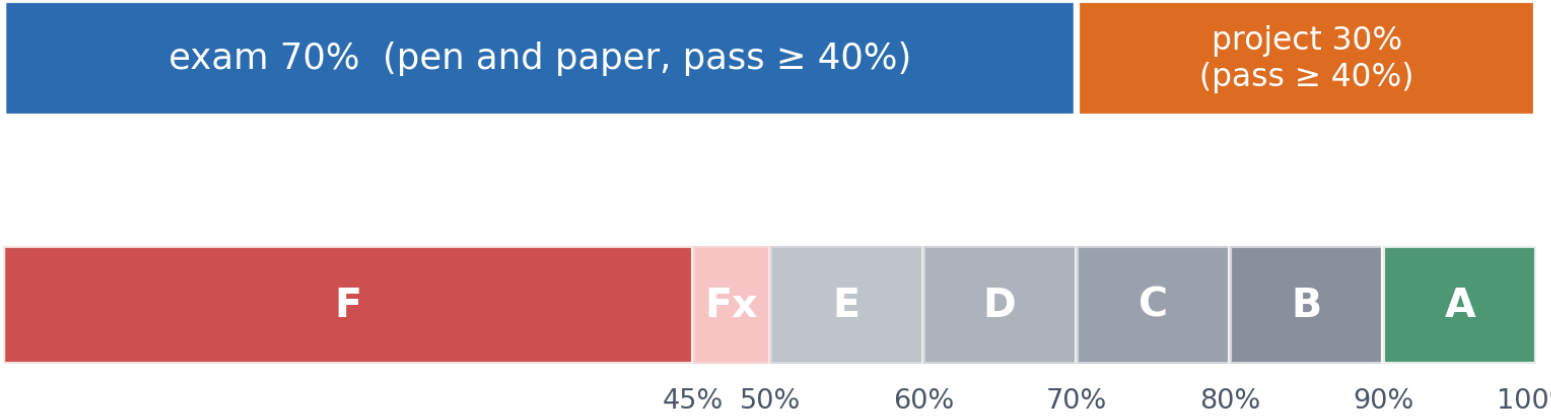
map (Google Maps)

Scan the code for the map. Directions: bioinfo.se/contact.

Address and directions from bioinfo.se/contact; rooms from the schedule. https://bioinfo.se/contact/

Labs, project, exam

+ all labs passed (each lab quiz ≥ 50%)



total score: 50% to pass

Labs: a short Canvas quiz and an anonymous feedback survey each, due the evening before the next lecture. Project: your own secondary-structure predictor (notebook, report, oral code presentation to a TA). Exam: pen and paper, up to five single-sided pages of your own notes. Bonus exam points, up to 3 each: questions sent in before the lectures, the live quiz in every lecture, and labs on time.

Details: course.bioinfo.se/course-info.html https://course.bioinfo.se/course-info.html

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Your work must be your own

Encouraged

talk and learn

- Discussing methods with other students
- AI tools as a 'science buddy'
- Asking when you are unsure, before you submit

Not allowed

plagiarism

- Submitting others' answers, code or text
- AI-written labs or reports
- AI tools at the exam

See course.bioinfo.se/course-info.html, 'Plagiarism' and 'AI tools'.

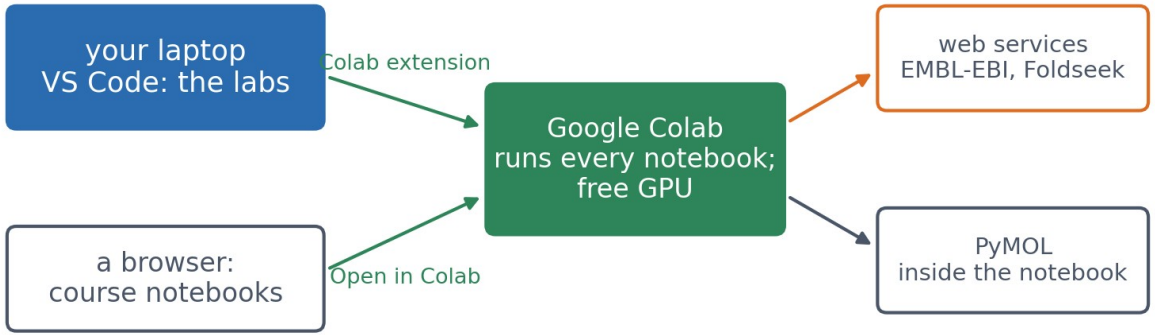
At SciLifeLab: the speed gates



- **Your card, every time, in and out**
- One person at a time
- A visitor: card twice, a few seconds apart
- **Never force your way through a gate**

Session 0 page, 'At SciLifeLab'. You need a SciLifeLab access card: when and where to collect it is on Canvas or by email.

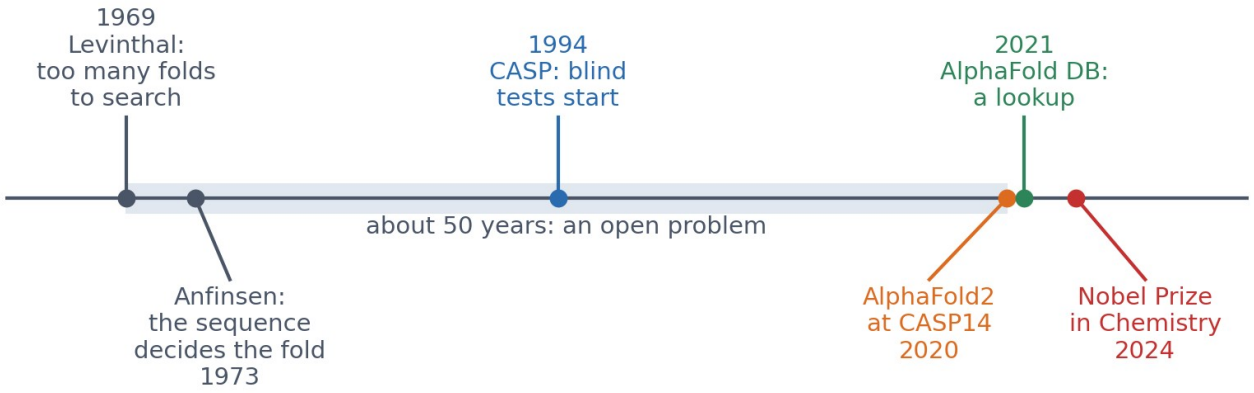
Computers



- **Every notebook runs on Colab**
- Labs: in VS Code, running on Colab
- **No suitable laptop? Tell us today**

Step by step: course.bioinfo.se/setup.html https://course.bioinfo.se/setup.html

From a 50-year problem to a database lookup

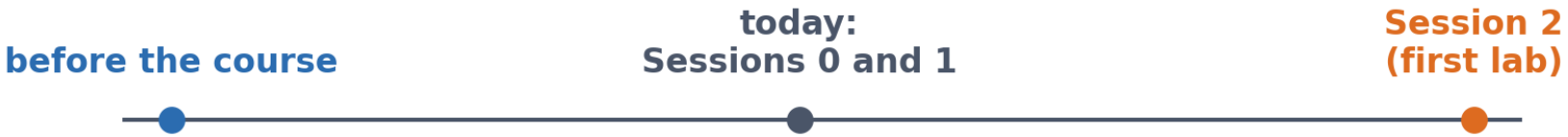


- 1960s-1970s: the fold is in the sequence
- 1994-2018: steady, slow progress at CASP
- 2020-2021: AlphaFold2, then a database
- **What will your field look like when you finish your PhD?**

Levinthal 1969; Anfinsen 1973, Science 181:223; predictioncenter.org (CASP); Jumper et al. 2021, Nature 596:583; nobelprize.org (2024).

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What was due before today, and before Session 2



1. Accept the Canvas invitation
2. Log in to quiz.bioinfo.se
3. Read the Session 1 page

(sent by email before the course)

Work through the Setup page

(about half an hour):

VS Code, git, Colab extension,

clone the course notebooks

Sessions 0 and 1 are this morning, so the three items on the left were emailed before the course started. Session 2 has the first lab, on Colab: sort out the setup (about half an hour) before it, not during it.

course.bioinfo.se/session00.html, 'Before the first lecture' and 'Before Session 2'.

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Session 1 next

Vectors and matrices, derivatives and gradients, probability, likelihood and entropy
Read the page before the lecture